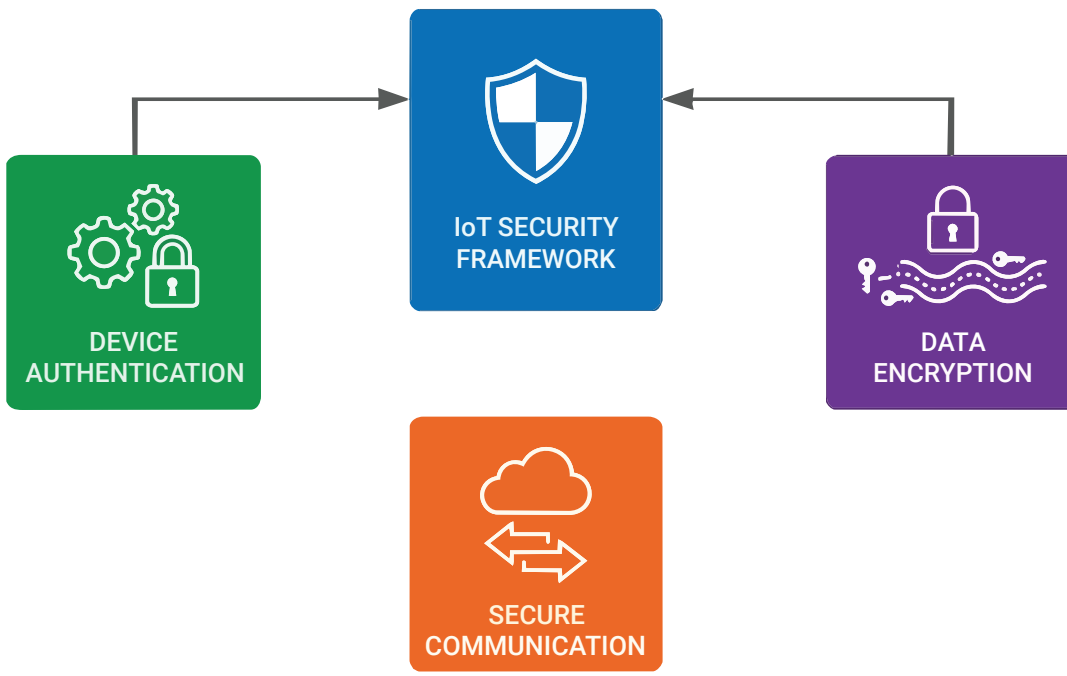
IoT Security Frameworks: Facilitating Device Authentication and Data Encryption

Open source IoT security frameworks are great at identifying devices and keeping the conversation between them private. It helps that their code is public and free to use.



Internet of Things (IoT) devices are everywhere: smart meters, factory sensors, cameras, door locks and wearables. Most of them are small, cheap and expected to run for years without anyone touching them. That combination makes them attractive targets. A single device with a default password or an unencrypted connection can give an attacker a way into an entire network.

Two questions sit at the heart of IoT security.

First, who is this device really talking to, and is it who it claims to be? That is authentication.

Second, even if the right parties are talking, can anyone else read or change what they say? That is encryption and integrity protection.

A good design answers both, from the moment the device boots to the moment its data reaches the cloud. Writing your own cryptography is a famously bad idea, so teams usually reuse existing code. Open source frameworks are popular for this because their code is public, widely reviewed, free to use and not tied to one vendor. They still need correct setup, though, and a well-known library used badly is not safe.

Security foundations for IoT

A useful starting point is the NIST baseline for IoT devices, which lists six core capabilities a device should have: device identification, device configuration, data protection, logical access to interfaces, software update, and cybersecurity state awareness. Authentication and encryption sit mainly under the first, third and fourth of these. Table 1 shows how common security goals map to open source mechanisms.

Security works best in layers, so that a failure in one place does not expose everything. Figure 1 shows a simple three-layer view that we will use throughout this document.